Engineering Mathematics

GATE CSE 2027 · complete formula, rule, method & trap sheet

36 reference boxes · **5** chapters · **18** worked examples (switch on above)

Linear Algebra | Calculus | Probability & Statistics | Problem → Standard Method Playbook | Last-minute one-liners

1 Linear Algebra

Determinant properties (n × n matrix) Core

FORMULA / RULE

```
det(AB) = det(A)·det(B)
det(Aᵀ) = det(A)
det(kA) = kⁿ·det(A)
det(A⁻¹) = 1/det(A)
Swapping two rows → sign changes
Two equal rows or a zero row → det = 0
Row op "add k·row i to row j" → unchanged
Triangular matrix: product of diagonal
```

FORMULA / RULE

Invertibility — all equivalent:

```
A is invertible
det(A) ≠ 0
rank(A) = n
Ax = 0 has only x = 0
columns are linearly independent
no eigenvalue is 0
```

Rank and systems of linear equations Core

RULES & FACTS

- **Rank** = number of non-zero rows in row-echelon form = dimension of the column space. rank(A) ≤ min(m, n).
- **Rank–nullity**: rank(A) + nullity(A) = n (number of columns). A 5×7 matrix of rank 3 has nullity **4**.
- Ax = b is **consistent** iff rank(A) = rank([A | b]). Unique solution iff the common rank = n (number of unknowns); infinitely many iff rank < n; none if the ranks differ.
- **Homogeneous** Ax = 0 always has the trivial solution; a square system has a non-trivial one iff **det(A) = 0**. If m < n there is always a non-trivial solution.

Eigenvalues and eigenvectors Core

NOTES

λ is an eigenvalue of A if Ax = λx for some x ≠ 0, i.e. det(A − λI) = 0 (the characteristic equation).

RULES & FACTS

- **Sum of eigenvalues = trace** (sum of diagonal); **product of eigenvalues = det(A)**.
- Triangular (and diagonal) matrices: eigenvalues are the diagonal entries.
- Eigenvalues of A^k are λ^k, of A^{-1} are $1/\lambda$, of A + cI are λ + c, of A^T are the same as A.
- **Cayley–Hamilton**: every matrix satisfies its own characteristic equation.
- Symmetric real matrices: all eigenvalues real, eigenvectors for distinct eigenvalues orthogonal. Eigenvectors for **distinct** eigenvalues are linearly independent; n distinct eigenvalues ⇒ diagonalisable.
- A singular matrix has 0 as an eigenvalue.

LU decomposition Core

NOTES

A square matrix (with non-zero pivots) factors as A = LU, with L lower-triangular (1s on the diagonal) and U upper-triangular. Solve Ax = b by forward substitution (Ly = b) then back substitution (Ux = y). Cost: about **$(2/3)n^3$** operations versus much more for computing the inverse.

2 Calculus

Standard limits Core

KEY POINT

```
lim (x→0) sin x / x = 1
lim (x→0) tan x / x = 1
lim (x→0) (1 − cos x)/x² = 1/2
lim (x→0) (eˣ − 1)/x = 1
lim (x→0) ln(1+x)/x = 1
```

KEY POINT

```
lim (x→∞) (1 + 1/x)ˣ = e
lim (x→∞) (1 + a/x)ˣ = eᵃ
lim (x→0) (1 + x)^(1/x) = e
lim (x→∞) xⁿ / eˣ = 0
lim (x→∞) ln x / x = 0
```

NOTES

L'Hôpital's rule: if f/g gives 0/0 or ∞/∞, then lim f/g = lim f'/g' (when the latter exists). Other indeterminate forms (0·∞, ∞−∞, 1^∞, 0^0, ∞^0) are first rewritten into a quotient. Taylor expansion also works: `eˣ = 1 + x + x²/2! + …`, `sin x = x − x³/3! + …`, `cos x = 1 − x²/2! + …`.

Continuity and differentiability Core

RULES & FACTS

- f is continuous at a if $\lim_{x \to a} f(x) = f(a)$. **Differentiable ⇒ continuous**; the converse is false.
- f(x) = |x| is continuous everywhere but **not differentiable at 0** (a corner); $f(x) = x^{1/3}$ is continuous but has an infinite slope at 0.
- **Intermediate value theorem**: a continuous f on [a,b] takes every value between f(a) and f(b). **Extreme value theorem**: a continuous f on a closed interval attains a maximum and a minimum.

Mean value theorems Core

RULES & FACTS

- **Rolle**: f continuous on [a,b], differentiable on (a,b), f(a) = f(b) ⇒ some c with f'(c) = 0.
- **Lagrange (MVT)**: some c ∈ (a,b) with f'(c) = (f(b) − f(a)) / (b − a).

Maxima and minima Core

NOTES

Critical points: f'(x) = 0 (or undefined). Second-derivative test: f'' > 0 → local min, f'' < 0 → local max, f'' = 0 → inconclusive. For the **absolute** max/min on a closed interval, compare critical values **and the endpoint values**.

Integration — Core

RULES & FACTS

- Fundamental theorem: $\int_a^b f = F(b) - F(a)$. By parts: $\int u\,dv = uv - \int v\,du$.
- Odd function over $[-a, a]$: integral = 0. Even function: $2\int_0^a$.
- Wallis: $\int_0^{\pi/2} \sin^n x\,dx$ for odd n = $(n-1)(n-3)\ldots 2 / (n(n-2)\ldots 3)$.
- Improper: $\int_1^\infty 1/x^p\,dx$ converges iff $p > 1$; $\int_0^1 1/x^p\,dx$ converges iff $p < 1$. $\int_0^\infty e^{-x}\,dx = 1$.

Series — Extension

RULES & FACTS

- **Geometric**: $\Sigma\, a \cdot r^n$ converges iff $|r| < 1$, to $a/(1-r)$. $\Sigma_{n\geq 0}\,(1/2)^n$ = **2**.
- **p-series**: $\Sigma\, 1/n^p$ converges iff $p > 1$ (so the harmonic series $\Sigma\, 1/n$ diverges).
- Ratio test: $L = \lim |a_{n+1}/a_n|$; converges if $L < 1$, diverges if $L > 1$, inconclusive at 1. A convergent series must have terms $\to 0$ (converse false).
- Alternating series with decreasing terms $\to 0$ converges (Leibniz).

L'Hôpital's rule and Taylor series — High-yield

RULES & FACTS

- **L'Hôpital:** if f/g gives 0/0 or ∞/∞ at a point, then lim f/g = lim f′/g′ provided the latter limit exists; repeat if the form persists. Rewrite $0\cdot\infty$ as a quotient and $\infty - \infty$ as a single fraction first.
- **Maclaurin series:** $e^x = 1 + x + x^2/2! + x^3/3! + \ldots$ · $\sin x = x - x^3/3! + x^5/5! - \ldots$ · $\cos x = 1 - x^2/2! + x^4/4! - \ldots$ · $\ln(1 + x) = x - x^2/2 + x^3/3 - \ldots$ $(|x| < 1)$ · $1/(1 - x) = 1 + x + x^2 + \ldots$ $(|x| < 1)$.

Improper integrals, Gamma and Beta — Extension

RULES & FACTS

- **p-test:** $\int_1^\infty x^{-p}\,dx$ converges iff **p > 1**; $\int_0^1 x^{-p}\,dx$ converges iff **p < 1**. $\int_0^\infty e^{-x}\,dx = 1$.
- **Gamma:** $\Gamma(n) = \int_0^\infty x^{n-1} e^{-x}\,dx$, with $\Gamma(n+1) = n\cdot\Gamma(n)$, so $\Gamma(n+1) = n!$ for integers, and $\Gamma(\tfrac{1}{2}) = \sqrt{\pi}$.
- **Beta:** $B(m, n) = \int_0^1 x^{m-1}(1-x)^{n-1}\,dx = \Gamma(m)\Gamma(n)/\Gamma(m+n)$. Example: $B(2, 3) = (1!\cdot 2!)/4!$ = **1/12**.

3 Probability & Statistics

Foundations — Core

RULES & FACTS

- $P(A \cup B) = P(A) + P(B) - P(A \cap B)$. $P(A^c) = 1 - P(A)$.
- **Conditional**: $P(A|B) = P(A \cap B)/P(B)$. **Independent** iff $P(A \cap B) = P(A)P(B)$. **Mutually exclusive** events with non-zero probability are **not** independent.
- **Total probability**: $P(B) = \Sigma\, P(B|A_i)P(A_i)$ over a partition. **Bayes**: $P(A_i|B) = P(B|A_i)P(A_i) / P(B)$.

Random variables — Core

RULES & FACTS

- **E[X]** = $\Sigma\, x\cdot p(x)$ (or $\int x\, f(x)\,dx$). Linearity always holds: $E[X + Y] = E[X] + E[Y]$, even for dependent X, Y.
- **Var(X)** = $E[X^2] - (E[X])^2$. $Var(aX + b) = a^2 Var(X)$. For **independent** X, Y: $Var(X + Y) = Var(X) + Var(Y)$.
- $Cov(X,Y) = E[XY] - E[X]E[Y]$; independent ⇒ Cov = 0 (converse false). Correlation $\rho \in [-1, 1]$.
- **Chebyshev**: $P(|X - \mu| \geq k\sigma) \leq 1/k^2$. **Markov**: $P(X \geq a) \leq E[X]/a$ for $X \geq 0$.
- Expected number of fixed points of a random permutation = **1** (for any n). Expected tosses of a fair coin until the first head = **2**.

Standard distributions — Core

REFERENCE TABLE

Distribution	Model	Mean	Variance
Bernoulli(p)	one trial	p	p(1−p)
Binomial(n, p)	successes in n trials; $P(k) = C(n,k)p^k(1-p)^{n-k}$	np	np(1−p)
Geometric(p)	trials until first success	1/p	$(1-p)/p^2$
Poisson(λ)	events in a fixed interval; $P(k) = e^{-\lambda}\lambda^k/k!$	λ	λ
Uniform(a, b)	continuous, equally likely	(a+b)/2	$(b-a)^2/12$
Exponential(λ)	waiting time; $f = \lambda e^{-\lambda x}$	1/λ	$1/\lambda^2$
Normal(μ, σ^2)	bell curve	μ	σ^2

RULES & FACTS

- Exponential is **memoryless**: $P(X > s + t \mid X > s) = P(X > t) = e^{-\lambda t}$. Geometric is the discrete memoryless one.
- Normal: about 68%, 95%, 99.7% of the mass lies within 1, 2, 3 standard deviations. The sum of independent normals is normal. **Central limit theorem**: the mean of many i.i.d. variables is approximately normal.
- Binomial(n, p) $\to$ Poisson(np) for large n, small p.

PMF, PDF and CDF — Core

REFERENCE TABLE

	Discrete (pmf p)	Continuous (pdf f)
Valid when	$p(x) \geq 0$ and $\Sigma\, p(x) = 1$	$f(x) \geq 0$ and $\int f(x)\,dx = 1$
Probability of an event	$P(X \in A) = \Sigma_{x\in A}\, p(x)$	$P(a \leq X \leq b) = \int_a^b f(x)\,dx$; **P(X = x) = 0** for every single point
CDF F(x) = P(X ≤ x)	Step function	Continuous, with f = dF/dx

NOTES

Every CDF is non-decreasing, runs from 0 to 1, and $P(a < X \leq b) = F(b) - F(a)$. A density can exceed 1 (only its integral must equal 1); a probability cannot.

Descriptive statistics — Core

NOTES

For a right-skewed distribution mean > median > mode; for a symmetric one they coincide. Sample variance divides by n − 1 (unbiased); population variance divides by n.

Joint and marginal distributions — High-yield

RULES & FACTS

- The **marginal** distribution comes from summing (or integrating) the joint over the other variable: $p_X(x) = \Sigma_y\, p(x, y)$.
- X and Y are **independent** iff $p(x, y) = p_X(x)\cdot p_Y(y)$ for **every** pair. One failing cell is enough to show dependence.

Normal distribution and z-scores Core

FORMULA / RULE

Standardise: if $X \sim N(\mu, \sigma^2)$ then $Z = (X - \mu)/\sigma \sim N(0, 1)$. Use $\Phi(z) = P(Z \le z)$ with the symmetry $\Phi(-z) = 1 - \Phi(z)$. Useful values: $\Phi(0) = 0.5$, $\Phi(1) = 0.8413$, $\Phi(1.5) = 0.9332$, $\Phi(1.96) = 0.9750$, $\Phi(2) = 0.9772$. The central 95% lies within $\pm 1.96\sigma$.

Hypergeometric distribution Extension

NOTES

Drawing n items **without replacement** from N items of which K are "successes": $P(X = k) = C(K, k) \cdot C(N - K, n - k) / C(N, n)$, with mean $n \cdot K/N$. Compare the binomial, which assumes **with** replacement (independent trials) and has the same mean $n \cdot p$ but a larger variance.

4 Problem → Standard Method Playbook

METHOD

Logic, Relations, Counting, Recurrences & Graphs

The problem → method tables for all discrete topics are kept in one place: Discrete Mathematics → Problem → Standard Method Guide.

METHOD

Linear Algebra Methods

Problem	Standard method	Shortcut / trap
System Ax=b	Row-reduce augmented matrix; compare ranks.	Unique iff rank(A)=rank([A|b])=number of variables.
Find eigenvalues	det(A−λI)=0.	For triangular matrices, eigenvalues are diagonal entries.
Find eigenvectors	Solve (A−λI)x=0 for each λ.	Multiplicity of eigenvalue ≠ automatically number of eigenvectors.
Use Cayley–Hamilton	Find characteristic polynomial p(λ), then substitute A: p(A)=0.	Useful for reducing high powers of A.
Determine invertibility	det(A)≠0 ⇔ A invertible ⇔ rank(A)=n.	For square A.

METHOD

Calculus

Pattern	Method	GATE cue
Maximum/minimum	Find critical points, test endpoints/boundary and compare values.	Don't stop after f′=0.
Improper/infinite series convergence	Choose comparison, ratio, root or integral test based on term structure.	Ratio/root are especially useful for factorials/exponentials.

METHOD

Probability & Statistics Methods

Pattern	Standard method	Formula / trap
Conditional probability	P(A|B)=P(A∩B)/P(B).	Condition changes the sample space.
Reverse conditional / diagnosis	Bayes theorem.	P(A|B)=P(B|A)P(A)/P(B).
Discrete expectation	E[X]=Σx·P(X=x).	E[g(X)]=Σg(x)P(X=x), not generally g(E[X]).
Variance	Var(X)=E[X²]−E[X]².	Var(aX+b)=a²Var(X).
Binomial	n independent Bernoulli trials, constant p; use C(n,k)p^k(1−p)^{n−k}.	E=np, Var=np(1−p).
Poisson	Count events over fixed interval with rate λ; P(X=k)=e^{−λ}λ^k/k!.	E=Var=λ.
Exponential	Continuous waiting time; f(x)=λe^{−λx}, x≥0.	Memoryless property.
Normal	Standardize Z=(X−μ)/σ.	Use symmetry around 0.
Joint → marginal	Sum/integrate joint distribution over the other variable.	Check support before integrating.
Independence	Discrete: p(x,y)=pX(x)pY(y); continuous: fXY=fXfY.	Zero covariance does not generally imply independence.

METHOD

🧮 Representative GATE-style Worked Problems

Problem 1 · Probability

Let X~Binomial(10,0.2). Find E[X] and Var(X).

Method: E=np and Var=np(1−p).

Answer: E[X]=2, Var(X)=1.6.

Problem 2 · Linear Algebra

If A is a 3×3 matrix with characteristic polynomial $\lambda^3 - 6\lambda^2 + 11\lambda - 6$, find $A^3 - 6A^2 + 11A$.

Method: Apply Cayley–Hamilton directly: p(A)=0.

Answer: 6I.

Problem 3 · Recurrence

T(n)=2T(n/2)+n. Determine the asymptotic order.

Method: Master theorem: a=2,b=2,n^{log₂2}=n and f(n)=n.

Answer: Θ(n log n).

Problem 4 · Graph

An undirected graph has 10 vertices and 9 edges and is connected. What can you conclude?

Method: A connected graph with n−1 edges is a tree.

Answer: It is a tree; hence every edge is a bridge.

Problem 5 · Probability

X is exponential with rate 2. Find P(X>3).

Method: Survival function P(X>t)=e^{−λt}.

Answer: e^{-6}.

METHOD

⚠ Scope note

Keep the main revision path aligned to the GATE CSE syllabus. Any enrichment outside the syllabus should be explicitly labelled optional rather than mixed into the core revision path.

5 Last-minute one-liners

LAST-MINUTE · FORMULAS & RULES

- For finite sets, $|A \cup B| = |A| + |B| - |A \cap B|$.
- For recurrence $T(n) = aT(n/b) + f(n)$, compare $f(n)$ with $n^{\log_b a}$ before applying Master theorem.
- For a square matrix, $\det(AB) = \det(A)\det(B)$; if $\det(A) \neq 0$, A is invertible.
- For an exponential rate λ distribution, $P(X > t) = e^{-\lambda t}$. Standardize normal variables with $Z = (X - \mu)/\sigma$.
- Standardise $Z = (X - \mu)/\sigma$, with $\Phi(-z) = 1 - \Phi(z)$; $\Phi(1) = 0.8413$, $\Phi(1.96) = 0.975$. Hypergeometric mean = nK/N.

LAST-MINUTE · MUST REMEMBER

- A proposition is either true or false; compound propositions are evaluated using logical connectives.
- Implication $P \rightarrow Q$ is false only when P is true and Q is false.
- A relation can be reflexive, symmetric, antisymmetric, and transitive independently; check each property from its definition.
- A function is injective if distinct inputs have distinct outputs; surjective if every codomain element is hit; bijective if both.
- A graph is bipartite iff it contains no odd cycle.
- For continuous random variables, probabilities are areas under the density; $P(X = x) = 0$ for a continuous distribution.
- CNF can be built as a conjunction of clauses; DNF as a disjunction of terms. Use truth-table rows when algebra becomes error-prone.
- **Cayley–Hamilton:** every square matrix satisfies its own characteristic polynomial; use it to reduce high powers of A.
- Convergence-test choice is pattern-driven: ratio/root for factorials and powers; comparison/integral for rational-like terms.
- **Generating functions:** independent choices multiply; the answer is a coefficient. A variable limited to 0…m contributes $1 + x + \ldots + x^m$.
- For a continuous variable $P(X = x) = 0$ and $f = dF/dx$; a density can exceed 1 but its integral is 1. Independence needs $p(x,y) = p_X(x) \cdot p_Y(y)$ in every cell.

LAST-MINUTE · COMMON TRAPS

- $P \rightarrow Q$ is logically equivalent to $\neg P \vee Q$, not $P \vee Q$.
- Mean, median, and mode are different statistics; do not infer one from another without additional distribution assumptions.
- Zero covariance does not generally imply independence. For independence, factor the joint distribution into marginals.
- **Free vs bound:** a quantifier covers only the formula right after it, so the same letter can be bound in one place and free in another. A formula with no free variables is a sentence.
- Equal degree sequences do not prove graph isomorphism (C6 vs two triangles). Self-complementary graphs have $n(n-1)/4$ edges, so $n \equiv 0$ or $1 \pmod 4$.

LAST-MINUTE · GATE FAVOURITES

- Equivalence relations are reflexive, symmetric, and transitive; they partition a set into equivalence classes.
- Know conditional probability, Bayes theorem, expectation, variance, and standard distributions used in GATE-style numericals.
- For a finite set, equivalence relations correspond exactly to partitions of the set.
- **For binomial X~Bin(n,p):** $E = np$ and $\text{Var} = np(1-p)$. For Poisson(λ): $E = \text{Var} = \lambda$.
- Use complement for 'at least one', inclusion–exclusion for overlapping cases, and pigeonhole for guaranteed repetition.